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# Edge AI & IoT for Resilient Smart Infrastructure

TSYP14 TECHNICAL CHALLENGE

## SCOPE & TOPIC

The challenge focuses on:

**Edge AI • IoT • Embedded Systems • Smart Sensing • Wireless Communication • RFID • Intelligent Systems**

Teams are invited to develop solutions for resilient infrastructure and real-world applications by combining sensing, intelligent processing, connectivity and real-time action.

## Challenge problem

Develop an intelligent connected system that addresses a real-world problem by:

**Collecting information → Processing it intelligently at the edge → Producing an appropriate response or action**

The solution should demonstrate meaningful intelligence and decision-making close to the source of the data.

## Challenge goal

Encourage students to design and demonstrate innovative Edge AI and IoT solutions that combine:

- Sensing
- Intelligent processing
- Connectivity
- Real-time action

to address practical real-world challenges.

## Challenge instruction

Teams must:

**1. IDENTIFY A REAL-WORLD PROBLEM**

Select a practical problem and define the proposed solution.

**2. DEVELOP AN EDGE AI / IoT SOLUTION**

The system must integrate: Sensing → Edge Processing → Connectivity → Response / Action

**3. IMPLEMENT EDGE INTELLIGENCE**

A meaningful processing capability must be implemented at the edge.

Cloud-only solutions are not sufficient.

**4. CHOOSE APPROPRIATE TECHNOLOGIES**

Teams are free to choose:

- Hardware
- Software
- Communication technologies
- AI/ML methods
- Embedded technologies
- Other appropriate technologies

**5. BUILD A FUNCTIONAL PROTOTYPE**

The final solution must be demonstrated through a functional prototype.

**6. DEMONSTRATE THE COMPLETE SYSTEM**

Teams must provide a live demonstration showing: Sensing → Intelligent Processing → Decision-Making → Response / Action

## Deliverables

- **TECHNICAL REPORT**

Maximum 4 pages

The report must be submitted anonymously and should briefly describe:

- Problem
- Proposed solution
- System architecture
- Main technologies
- Intelligent processing approach
- Innovation
- Expected impact

- **FUNCTIONAL PROTOTYPE**

A functional:

- Hardware prototype
- Software prototype
- Hybrid prototype

- **PRESENTATION**

5-minute presentation

*The presentation must remain anonymous until the evaluation process is completed.*

- **LIVE DEMONSTRATION**

A live demonstration of the functional prototype is mandatory.

Teams should demonstrate:

**Sensing → Intelligent Processing → Decision → Resulting Action**

## Anonymity requirement

All submitted materials must remain anonymous during submission and evaluation.

Do not include:

- Team names
- Member names
- University names
- IEEE Student Branch affiliations
- Logos
- Email addresses
- Other identifying information

Any violation may result in immediate **disqualification**.

## Scoring

**TOTAL SCORE: 100 POINTS**

**FINAL SCORING SYSTEM — 100 POINTS**

**PHASE 1: Technical Proposal & Video Demonstration:**

Non-Technical: 6 pts

- Problem Relevance & Impact: 3 pts
- Innovation, Creativity & Originality: 3 pts

Technical: 8 pts

- Quality of the Proposed Solution: 2 pts
- Technical Implementation & Edge AI/IoT Integration: 3 pts
- Functional Prototype & Video Demonstration: 2 pts
- CASS/CRFID Integration & Deployment Potential: 1 pt

**Total: 14 pts**

**PHASE 2: Final Evaluation:**

Non-Technical: 2.5 pts

- Quality of Pitching & Presentation: 2 pts
- Final Defense & Compliance with Presentation Requirements: 0.5 pt

Technical: 3.5 pts

- Live Prototype Demonstration: 2 pts
- Technical Questions & Answers: 1.5 pts

**Total: 6 pts**

## Pitching

English

5 min pitching

## Team Requirements

Maximum team size: **3 members**

## Important dates

- Challenge Infosession: **18/09/2026**
- Phase 1 Submission Deadline: **20/11/2026**
- Final Submission Deadline: **09/12/2026**

## Winners

**3 WINNING TEAMS**

All winners receive:

- Official IEEE certificates
- Gifts / prizes
- Recognition during TSYP14

## Submission

***Submit your project and show us what you've built.***

## Contact



Please contact us through the following email address wassim.jmal@ieee.org for any inquires or clarifications.